Cosmology. The equation of state for the neutron pressure, however, was a more difficult issue, since the pressure would be influenced by the nuclear force (with which neutrons push and pull on each other). Although the nuclear force was becoming well understood at the densities inside atomic nuclei, it was very poorly understood at the ten times higher higher densities that neutrons would face deep inside a massive neutron star. Physicists did not even know whether the nuclear force was attractive at these densities or repulsive (whether neutrons pulled on each other or pushed), and thus there was no way to know whether the nuclear force reduced the neutrons' pressure or increased it. But Oppenheimer had a strategy to deal with these unknowns.

Pretend, at first, that the nuclear force doesn't exist, Oppenheimer suggested to Volkoff. Then all the pressure will be of a sort that is well understood; it will be neutron degeneracy pressure (pressure produced by the neutrons' "claustrophobic" motions). Balance this neutron degeneracy pressure against gravity, and from the balance, calculate the structures and masses that neutron stars would have in a universe without any nuclear force. Then, afterward, try to estimate how the stars' structures and masses will change if, in our real Universe, the nuclear force behaves in this, that, or some other way.

With such well-posed instructions it was hard to miss. It took only a few days for Volkoff, guided by daily discussions with Oppenheimer and by Tolman's book, to derive the general relativistic description of gravity inside a neutron star. And it took only a few days for him to translate the well-known equation of state for degenerate electron pressure into one for degenerate neutron pressure. By balancing the pressure against the gravity, Volkoff obtained a complicated differential equation whose solution would tell him the star's internal structure. Then he was stymied. Try as he might, Volkoff could not solve his differential equation to get a formula for the star's structure; so, like Chandrasekhar with white dwarfs, he was forced to solve his equation numerically. Just as Chandrasekhar had spent many days in 1934 punching the buttons of Eddington's Braunschweiger calculator to compute the analogous white-dwarf structure, so Volkoff labored through much of November and December 1938, punching the buttons of a Marchant calculator.

While Volkoff punched buttons in Berkeley, Richard Tolman in Pasadena was taking a different tack: He strongly preferred to express the